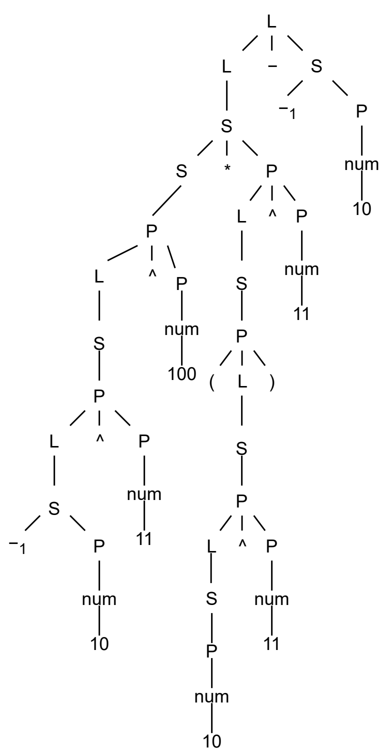
1a.) An unambiguous grammar that generates L.

G = (V, Σ, R, S) such that
V = {+, -, *, /, $-_1$, ^, (,), L, S, P, num}
Σ = {+, -, *, /, $-_1$, ^, (,), num}
R = {L → L + S,
L → L - S,
L → S,
S → $-_1$P,
S → S * P,
S → S / P,
S → P,
P → L ^ P,
P → (L),
P → num},
S = L

This unambiguous grammar, G, is correct and generates L because G creates only one parse tree for each input string, w into the language generated by G, L(G). Because a grammar is considered ambiguous if and only if there is at least one string in L(G) for which G produces more than one tree, then this grammar G would be considered unambiguous as it can only produce one parse tree for every string input into the language.

1b.) the parse tree that your grammar produces for the string : $-_1$10^11^100 ∗ (10^11)^100 − $-_1$10



2.) Because ⌈x^n⌉ will always evaluate to a positive number from any real number x, then the only time the L_x cannot be generated from a context free grammar for x^n is when (m = x^n) m is equal to a negative value (resulting in a $1/a^m$ value). If m were positive then L_x can be defined by grammar G, For m to be positive, x must be any real number greater than 0.

$L_x = \{a^{x\wedge n}, n \geq 0\}$

For $x = k^n$ we have $w = a^{k\wedge 2n} = uvxyz$

$vy = a^p$ for some nonzero p (because $n \geq 0$)

For q = n, $a^{k\wedge(2n+n)}$ becomes $a^{k\wedge 3n}$ and must be in L

For all q values, $|vxy| \leq k$ so therefore for all positive values of x,

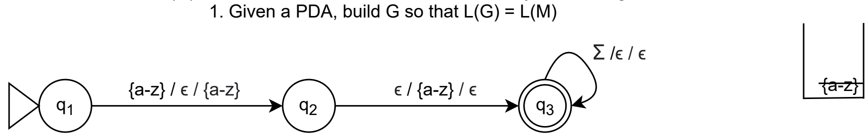
$vy \neq \epsilon$, $|vxy| \leq k$, and $\forall q \geq 0$, $uv^qxy^qz \in L_x$

and is a context free language.

3.) Given a context-free grammar G, does L(G) contain only alphanumeric strings that start with a letter? (decidability)

To determine if the L(G) is decidable, we will follow the decidability based on grammar method.

1. Given a PDA, build G so that L(G) = L(M)



G = (V, Σ, R, S) such that
V = {S, R, {a-z}, {0-9}}
Σ = {{a-z}, {0-9}}
R = {S -> {a-z}P
P -> {a-z}P | {a-z},
P -> {0-9}P | {0-9},
P -> ϵ },
S = S

2. A string is not nullable because w ≠ ϵ so G needs to be rebuilt as G' in Chomsky normal form such that L(G') = L(G) - {ϵ}. To do that we will remove all ϵ rules and then any unit productions, then the mixed rules, and then any long removes from the grammar G.

- Removing ϵ rules: S -> {a-z}P | {a-z} | {0-9}P | {0-9}
- P -> {a-z}P | {a-z} | {0-9}P | {0-9}
- No unit productions to remove
- Removing mixed rules: Where {a-z} mixed rules will become T_a and {0-9} mixed rules will become T_0
- S -> T_aP | {a-z} | T_0}P | {0-9}
- P -> T_aP | {a-z} | T_0P | {0-9}
- T_a -> {a-z}
- T_0 -> {0-9}
- No long removes needed

Running any w through L(G') will derive w, therefore we can confirm the context free language using the grammar.

Because membership (given a language L and a string w, is w in L?) can be be confirmed if L is context free, then there exists some context free grammar, G that generates it and there also exists some PDA that accepts it as seen above.

G can derive w, and will do so in 2|w| - 1 steps, because for each w, a letter needs to start it which will be put onto the stack and then popped off as the grammar goes to its accepting state, which can happen after one step.

Therefore, the problem of "Given a context-free grammar, G, does L(G) contain only alphanumeric strings that start with a letter" is decidable.